AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-01 | Entry window 1 November 2026 (full day) | 7 records screened in scope

7	6	6	1	2
RECORDS IN SCOPE (10 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

China, 466,543 kidney deaths: coarse $PM_{2.5–10}$ per IQR (OR 1.024, 1.016–1.032) is comparable to $PM_{2.5}$ (OR 1.020, 1.012–1.027), with renal failure the most affected subtype.

Nationwide time-stratified case-crossover, 2013–2019, modelled daily concentrations at the residential address. NO_2 was the strongest of the four pollutants per IQR (+3.62 %, 2.47–4.78); renal-failure and glomerular deaths responded to all four, tubulointerstitial to O_3 only, urolithiasis to none. **Where it is weak:** the abstract gives no IQR widths, so the pollutant ranking is not per unit mass; lags differ by pollutant; renal failure on a death certificate is often a proximate cause. **Why it matters here:** a coarse-mode association this size sits in the size range where low-cost optical nodes are least reliable.

What changed today

- **Accra schoolchildren: no pro-hypertensive signal.** 919 children, home $PM_{2.5}$ SBP −0.70 mmHg (−3.34 to 1.94) per 10 units, in a city at 5–8 times the WHO annual guideline. Inverse point estimates suggest confounding. LUR built on a 146-site campaign. BACKFILL, entry 4 June.
- **Low-level mortality gradient shrinks under geography.** Native American Medicare beneficiaries OR 1.12 (1.06–1.18) per $\mu g\,m^{-3}$ vs 1.01 (1.001–1.02); most attenuation came from state-level clustering. BACKFILL, entry 4 May.
- **SPIROMICS proteomics.** 860 plasma proteins at FDR<0.05 for prior-year $PM_{2.5}$, none for NO_2 (n≈1,620, congress abstract, no effect sizes). BACKFILL, Crossref deposit 29 October.
- **Ferrimagnetic nanoparticles in children's brains.** Mexico City autopsy tissue, 12 children and 16 adults; presence and field-driven motion shown, no comparison group, no effect size.
- **Sunday floor.** PubMed entered 5 candidate records; the day's Crossref yield was Elsevier deposits without abstracts. Two records ride on metadata alone (German sick-leave claims, personal air purifier); retry the *Environ Int* abstract first.
- **Exposure basis (x1).** Of the four health records with a recoverable basis, three are modelled (national grid, LUR, prior-year surface) and one is a laboratory tissue assay: nothing measured on a person and no low-cost-sensor record today.

Window continuity

The 31 October issue closed with `last_entry_date` at 31 October. This issue was built on the **2 November 05:0x UTC (1 November 23:0x CST) run**; the 22:00 rule is satisfied, so 1 November is complete. It covers **1 November in full** (Sunday, first day of standard time). One record (Calderón-Garcidueñas, EDAT 31 October) surfaced only on Europe PMC's 1 November leg and was not screened on 31 October, so it is assigned here. Three BACKFILL records are carried, the working cap, all marked with their entry dates. No rollup is owed: W45 (1–7 November) falls on Saturday 7 November.

Corpus analytics

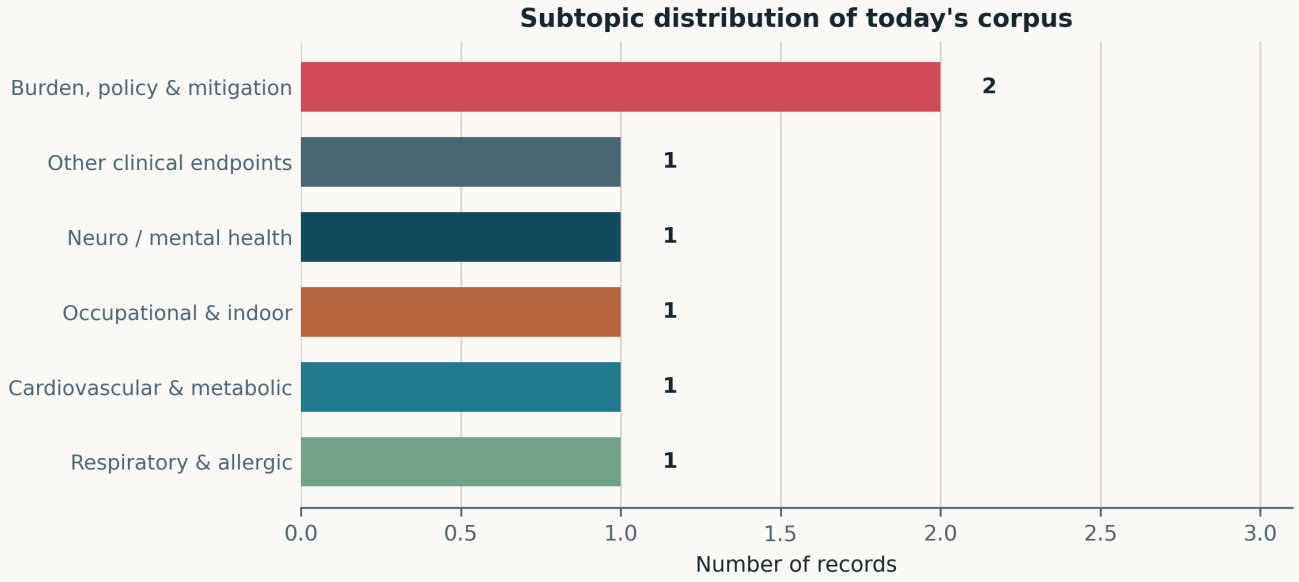


Fig. Subtopic assignment of the 7 records across six bands: burden and policy two (sick leave, Medicare mortality); respiratory, cardiovascular, neuro, occupational & indoor and other clinical one each. Mechanistic, reproductive, exposure and sensing bands are empty.

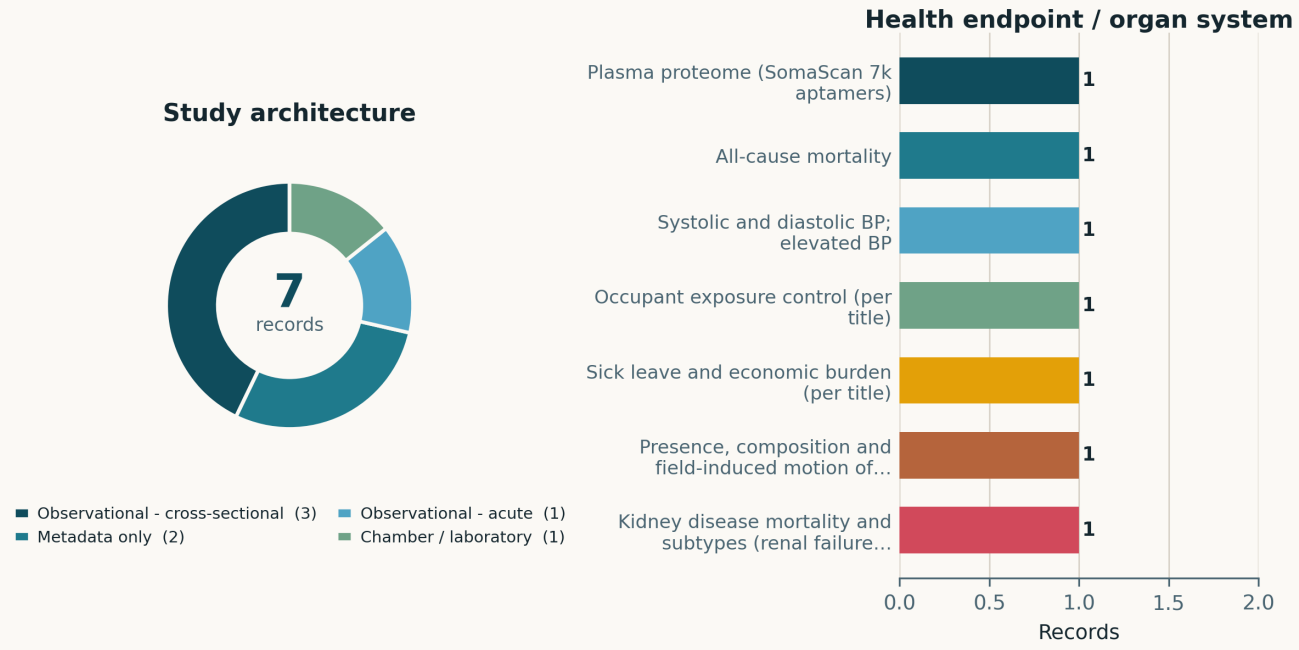


Fig. Left — architecture: observational cross-sectional three (Accra, Medicare case-control, SPIROMICS baseline), metadata-only two, and one each acute case-crossover and laboratory (autopsy tissue). Right — endpoints from kidney mortality and blood pressure to the plasma proteome and sick leave.

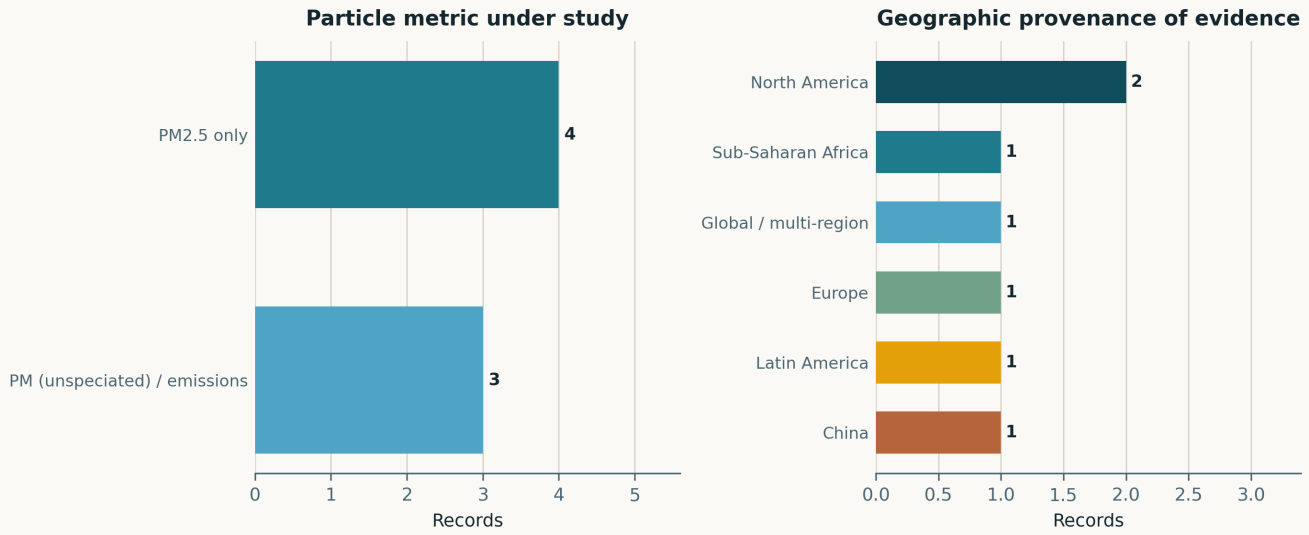


Fig. Left — particle metric: $PM_{2.5}$-only 4, unspeciated 3. Right — geography: North America 2, then one each Sub-Saharan Africa, Europe, Latin America and China; the personal-purifier record states no site and is drawn under global/multi-region.

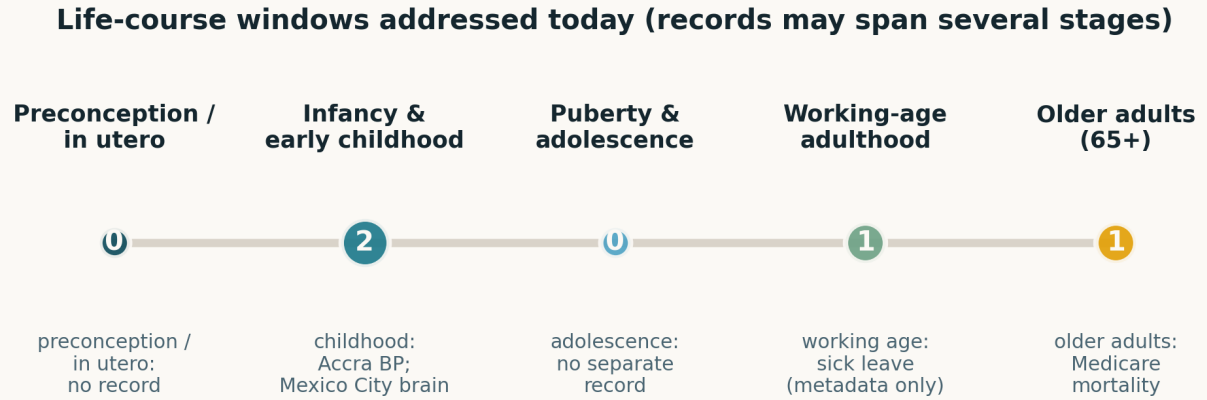


Fig. Life-course windows: two childhood records (Accra blood pressure, Mexico City brain tissue), one working-age (sick leave, metadata only) and one older-adult (Medicare mortality). No in-utero or separate adolescent record.

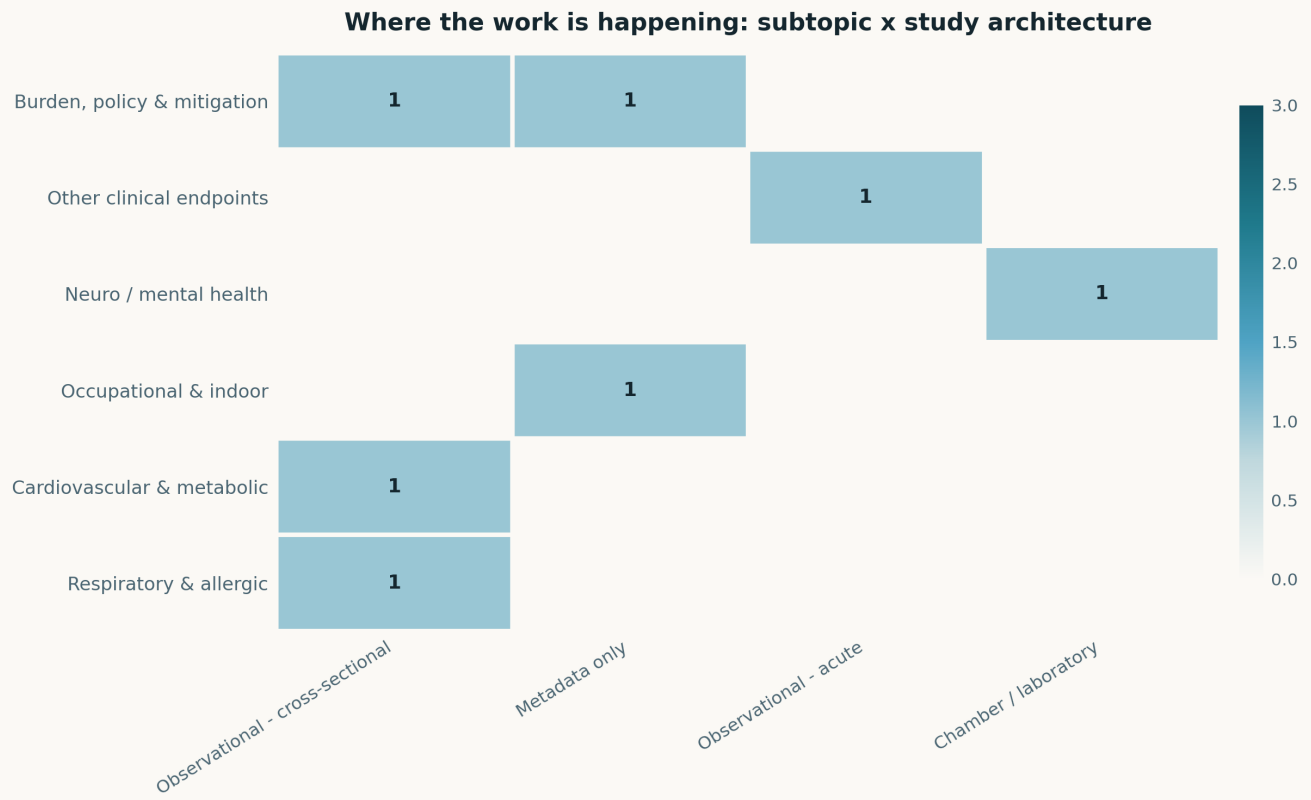


Fig. Subtopic × study architecture. Seven records occupy seven cells; no cell has a replicate today, so this is a map of what arrived, not a pattern.

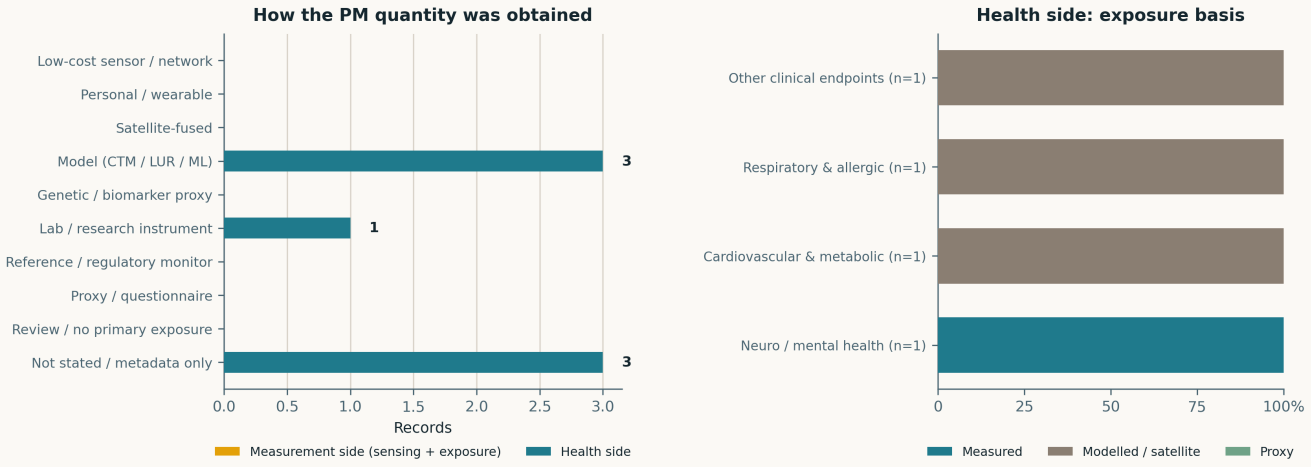


Fig. How each record obtained its PM quantity, classified from the exposure field. 0 of 7 records rest on a low-cost sensor or sensor network; 0 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=4): measured 25 %, modelled or satellite-fused 75 %, proxy 0 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

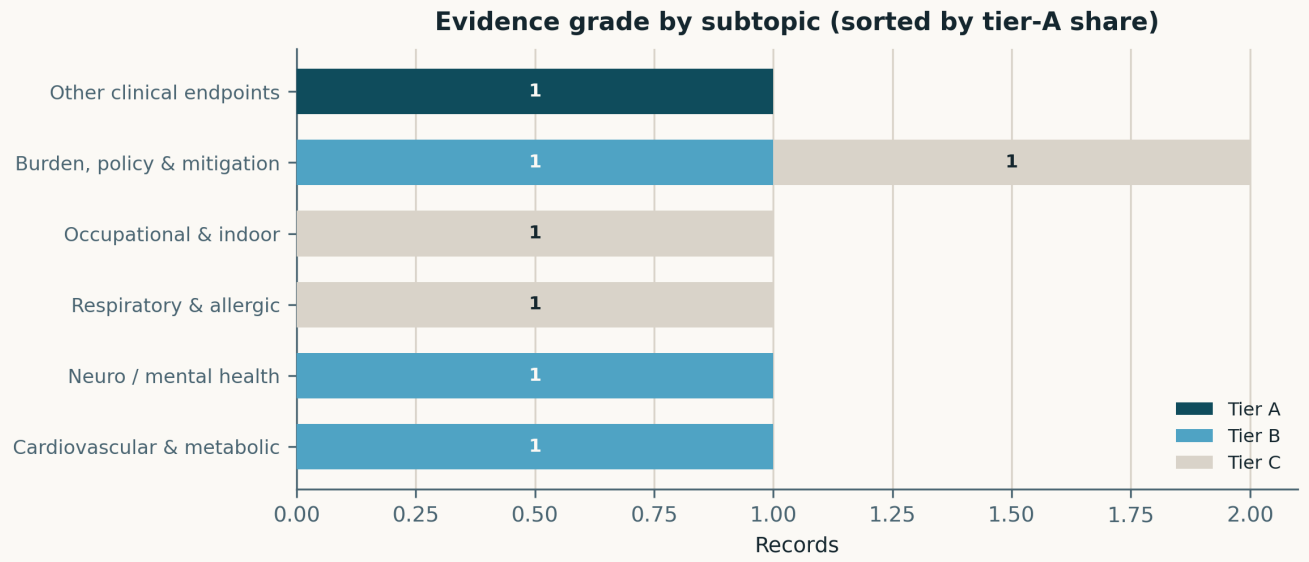


Fig. Tier mix per subtopic. 1 of 7 records are tier A (14 %). Highest tier-A share: *Other clinical endpoints* (1 of 1).

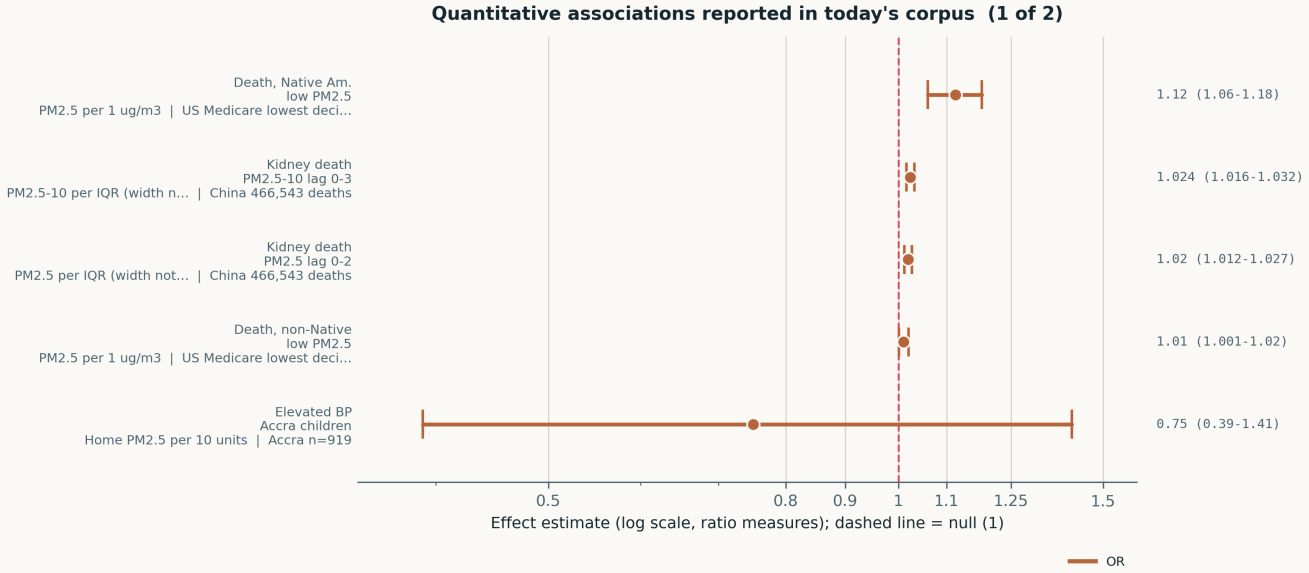


Fig. Ratio scale, five estimates from three studies. Increments differ (per IQR with undisclosed widths in China; per 1 $\mu g\,m^{-3}$ in Medicare; per 10 units in Accra) and the Medicare slope is a low-decile contrast: nothing here is comparable across studies or poolable.

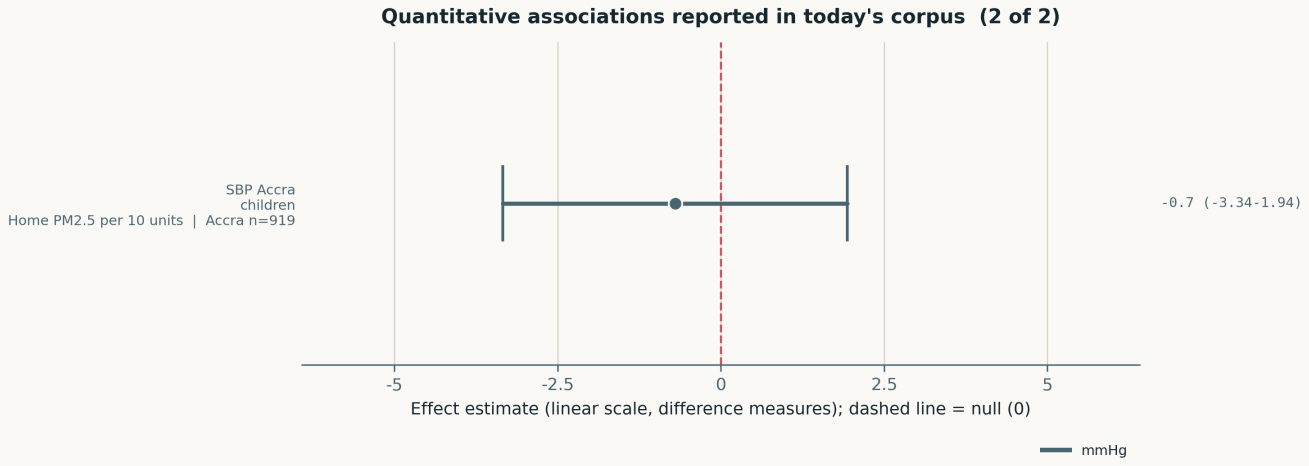


Fig. Difference scale, one estimate: systolic blood pressure per 10 units of home $PM_{2.5}$, Accra schoolchildren (n=919), an inverse point estimate with a CI spanning zero.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Calderon-Garciduenas ferrimagnetic nanoparticles olfactory bulb cortex Mexico City children 2026 Gac Med Mex

The olfactory bulb, prefrontal and anterior cingulate cortex are early pediatric targets for anthropogenic magnetic nanoparticles and their motion behavior under electromagnetic fields.

48 brain samples (olfactory bulb, anterior cingulate, prefrontal cortex) from 12 children (8.5 +/- 7.6 y) and 16 adults (39.2 +/- 19.6 y), Metropolitan Mexico City. Magnetophoresis at 25-1,000 mT AC showed small displacements and significant vector-direction variation; TEM and EDX located ferrimagnetic nanoparticles. No effect size. Limitation: autopsy convenience series with no comparison population or exposure measure in the abstract, so 'early pediatric targets' and the neural-damage risk are inferred, not tested; particle attribution to anthropogenic sources is not apportioned here. Relevance: low for sensing - mass-calibrated optical nodes are blind to this metallic ultrafine fraction, which is the actual argument for co-located number or magnetic measurements.

doi: 10.24875/gmm.m26001158

Cardiovascular & metabolic — 1 record

Alli ambient air pollution blood pressure schoolchildren Accra LUR 2026 Sci Rep

Urban ambient air pollution and blood pressure in schoolchildren in Accra, Ghana

BACKFILL (entry date 4 Jun 2026; Amass). 919 children aged 7-14 in 90 schools; one-third with elevated BP. Per 10 units home exposure: SBP $PM_{2.5}$ -0.70 mmHg (-3.34 to 1.94), BC -4.61 (-9.59 to 0.37); DBP $PM_{2.5}$ -0.83 (-2.93 to 1.28), BC -4.35 (-8.19 to -0.50); elevated BP $PM_{2.5}$ OR 0.75 (0.39-1.41). Exposures ran 5-8 times the WHO annual $PM_{2.5}$ guideline. Limitation: cross-sectional, and inverse associations in a high-exposure city suggest residual confounding by socioeconomic or adiposity patterning; the upper bound does not exclude a moderate SBP rise. Relevance: high - the LUR rests on a 146-site campaign in sub-Saharan Africa, the design a low-cost network could sustain.

doi: 10.1038/s41598-026-54135-6

Respiratory & allergic — 1 record

Ruvuna ambient $PM_{2.5}$ NO_2 plasma proteomics SPIROMICS SomaScan congress 2026 Eur Respir J (congress abstract)

Associations of ambient air pollution and protein expression among individuals from the Subpopulations and intermediate outcome measures in COPD study (SPIROMICS)

BACKFILL (Crossref deposit 29 Oct 2026; Amass; no PMID). ERS congress abstract. SPIROMICS tobacco-exposed cohort, n=1,627 for $PM_{2.5}$ and n=1,618 for NO_2; SomaScan 7k against modelled prior-year exposure, adjusted for site, COPD status, smoking, BMI, occupation, income and deprivation. 860 proteins associated with $PM_{2.5}$ at FDR<0.05, none with NO_2; enrichment in vascular remodelling, insulin signalling and inflammation. Limitation: abstract only, no effect sizes or replication, cross-sectional, and a one-pollutant-sided result may partly reflect exposure-model quality or spatial confounding. Relevance: low - no sensing element, but the proteomic contrast depends on exposure accuracy.

doi: 10.1183/13993003.congress-2026.oa5535

Occupational & indoor — 1 record

Nie personal air purification device multi-scale modelling fan speed occupant exposure 2026 Build Environ

Multi-scale modeling and fan speed optimization of a personal air-purification device for occupant exposure control in polluted indoor environments

METADATA ONLY. Metadata only: Elsevier day-0 deposit, no abstract in Crossref or Semantic Scholar; no finding asserted. The title describes multi-scale modelling and fan-speed optimisation of a personal air-purification device. Limitation: the title does not confirm particulate matter as the pollutant, and no performance figures, validation or occupant data are available. Relevance: to be assessed - breathing-zone control is where stationary sensors misstate exposure; retain only if the abstract confirms a PM target and measured validation.

doi: 10.1016/j.buildenv.2026.115523

Burden, policy & mitigation — 2 records

Aretz sick leave long-term fine particulate German health claims 13-year 2026 Environ Int

Assessing the health and economic burden of sick leave associated with long-term fine particulate matter exposure: a 13-year analysis using German health claims data

METADATA ONLY. Metadata only: Elsevier day-0 deposit, no abstract in Crossref, Europe PMC or Semantic Scholar; no finding asserted. The title describes a 13-year analysis of German health claims linking long-term fine PM to sick leave and its economic burden. Limitation: exposure source, increment, cohort size and estimate are all unknown until the abstract deposits. Relevance: to be assessed - claims-linked sick leave is a working-age endpoint rarely paired with PM, and this is the first record to retry on the next weekly abstract sweep.

doi: 10.1016/j.envint.2026.110586

Hess low-level $PM_{2.5}$ mortality Native American Medicare geographic attenuation 2026 Int J Environ Res Public Health

Untangling Impacts of Socioeconomic Position, Chronic Disease, and Low-Level $PM_{2.5}$ Exposure on Mortality Among Native American Medicare Beneficiaries

BACKFILL (entry date 4 May 2026; Amass). Case-control within 1,713,399 traditional Medicare beneficiaries in the lowest $PM_{2.5}$ decile, 2015-2016, with PCA summaries of area SEP and chronic disease. Fully adjusted OR per $\mu g\,m^{-3}$: Native American 1.12 (1.06-1.18), non-Native American 1.01 (1.001-1.02). Sequential adjustment attributed most attenuation in Native American beneficiaries to state-level clustering, less to SEP and disease. Limitation: the abstract gives no pre-adjustment coefficient, zip-level exposure over a narrow range makes a per-$\mu g\,m^{-3}$ slope fragile, and state effects may absorb real exposure contrast. Relevance: moderate - low-level gradients are where sparse monitoring error is largest.

doi: 10.3390/ijerph23040464

Other clinical endpoints 1 record

Xiao air pollution kidney disease mortality subtypes China case-crossover 2026 Chin Med J (Engl)

Air pollution exposure and death from different subtypes of kidney diseases: A nationwide, individual-level, case-crossover study in China.

466,543 kidney deaths, China 2013-2019, time-stratified case-crossover with conditional logistic regression. Per IQR: $PM_{2.5}$ lag 0-2 OR 1.0195 (1.0122-1.0269); coarse $PM_{2.5}$-10 lag 0-3 OR 1.0240 (1.0158-1.0323); O_3 +2.05% (1.05-3.07); NO_2 lag 0-3 +3.62% (2.47-4.78). Renal failure and glomerular deaths responded to all four pollutants; tubulointerstitial deaths to O_3 only; urolithiasis null. Limitation: the abstract does not give the IQR widths, so 'NO_2 strongest' is not a per-mass comparison, the lag differs by pollutant, and renal failure on a death certificate is often a proximate cause. Relevance: moderate - coarse-mode signal comparable to $PM_{2.5}$ per IQR, in a size fraction low-cost optical sensors undercount.

doi: 10.1097/cm9.0000000000004318

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Calderon-Garciduenas ferrimagnetic nanoparticles olfactory bulb cortex Mexico City children	The olfactory bulb, prefrontal and anterior cingulate cortex are early pediatric targets for anthropogenic magnetic nanoparticles and their motion behavior under electromagnetic fields.	*Gac Med Mex*	Autopsy tissue analysis	Ferrimagnetic nanoparticles in brain tissue by magnetophoresis, TEM and EDX	Mexico (Mexico City)
Cardiovascular & metabolic					
Alli ambient air pollution blood pressure schoolchildren Accra LUR	Urban ambient air pollution and blood pressure in schoolchildren in Accra, Ghana	*Sci Rep*	School-based cross-sectional	Home and school $PM_{2.5}$, BC, NO2 from spatiotemporal LUR (146 measurement sites)	Ghana (Accra)
Respiratory & allergic					
Ruvuna ambient $PM_{2.5}$ NO2 plasma proteomics SPIROMICS SomaScan congress	Associations of ambient air pollution and protein expression among individuals from the Subpopulations and intermediate outcome measures in COPD study	*Eur Respir J (congress abstract)*	Cross-sectional (cohort baseline)	Modelled prior-year average $PM_{2.5}$ and NO2	United States (SPIROMICS)
Occupational & indoor					
Nie personal air purification device multi-scale modelling fan speed occupant exposure	Multi-scale modeling and fan speed optimization of a personal air-purification device for occupant exposure control in polluted indoor environments	*Build Environ*	Metadata only (no abstract)	Not stated (indoor pollutant class unavailable)	Not stated
Burden, policy & mitigation					
Aretz sick leave long-term fine particulate German health claims 13-year	Assessing the health and economic burden of sick leave associated with long-term fine particulate matter exposure: a 13-year analysis using German health claims data	*Environ Int*	Metadata only (no abstract)	Long-term fine PM (exposure source not available)	Germany
Hess low-level $PM_{2.5}$ mortality Native American Medicare geographic attenuation	Untangling Impacts of Socioeconomic Position, Chronic Disease, and Low-Level $PM_{2.5}$ Exposure on Mortality Among Native American Medicare Beneficiaries	*Int J Environ Res Public Health*	Case-control	Long-term low-level $PM_{2.5}$ (lowest-decile zip codes; exposure source not stated)	United States (Medicare)
Other clinical endpoints					
Xiao air pollution kidney disease mortality subtypes China case-crossover	Air pollution exposure and death from different subtypes of kidney diseases: A nationwide, individual-level, case-crossover study in China.	*Chin Med J (Engl)*	Time-stratified case-crossover	Modelled daily $PM_{2.5}$ and $PM_{2.5}$-10 at residential address, with O3 and NO2	China (nationwide)

Provenance and method

Entry window **1 November 2026**, single day (Sunday). Harvester legs: **PubMed** health **0**, sensing **1**; **Europe PMC 9** after dropping 2 AGRICOLA retro-index records; **Crossref by ISSN 6**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429 (the OpenAlex connector is not on this session). 14 unique, 6 already carried, 8 fresh; PubMed connector (EDAT 1 Nov, three axes) 5 PMIDs, 1 seen, 4 appended (**12 screened**). Abstracts: 4 recovered by efetch; Elsevier day-0 deposits remain empty. **Amass** BiomedCore (ingest date filter) surfaced **3 backfill** records admitted and 2 rejected; others held. **Consensus** calibration sweep: 10 hits, 0 new (Fang 2026 already shipped; Feng 2026 review already rejected). **medRxiv** toolbox leg: no keyword filter, no PM record. bioRxiv/medRxiv and alphaXiv connectors are not on this session. **Scite** retraction check unavailable (connector requires a paid plan); Amass reports no retraction on PMID-bearing backfills. **Wiley Scholar Gateway** not called (needs re-authorisation). **7 admitted** (4 window, 3 backfill), **10 rejected** with logged reasons. **Trial watch** 1–2 November: ClinicalTrials.gov 0; Amass TrialCore 10 returned, one relevant (NCT07311967, UIC cooking UFP chamber, not previously logged). **2 carried on metadata alone**. Abstracts remain the copyright of their publishers.